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Lab 3: Polishing an Irregular Surface

Application B: Force Control with UR3e

Task Description

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The objective of this laboratory session was to program the Universal Robots UR3e collaborative robot to perform a continuous manufacturing task: polishing an irregular surface.

Unlike standard "Pick and Place" operations which rely on precise positional coordinates, polishing a curved surface (such as a car hood or wooden furniture) requires **Force Control**. The robot must maintain contact with the surface while adapting to changes in geometry without losing contact or applying excessive pressure.

This report details the implementation of force sensing, trajectory monitoring, and safety logic using the PolyScope programming interface.

Program Breakdown

The application logic is structured into three distinct phases: approach, surface detection, and force-controlled execution. The following pseudo-code illustrates the logic flow, including the parallel thread used for safety monitoring.

Algorithm 1 UR3e Car Polishing Logic (Application B)

Initialize System
Set TCP $(0, 0, 0)$ and Payload
Set Gravity Vector $[0, 0, 9.82]$
Phase 1: Approach Surface
MoveL(Waypoint_1) ▷ Fast approach
MoveL(Waypoint_4) ▷ Pre-contact position
Phase 2: Surface Detection
Start Thread: Move in Direction $Base_{Z-}$ ($v = 0.02$ m/s)
repeat
    Read Torque/Force Sensors
until `tool_contact_detection` is True
Stop Motion
Retract slightly to release tension
Phase 3: Polishing (Force Control)
zero_ftsensor ▷ Tare sensors
Enter Force Mode:
  Frame: Tool Frame
  Compliance: Z-axis enabled (1)
  Force Target: 10.0 Newtons ($Z+$)
  Limits: Speed < 100 mm/s
▷ Execute Zig-Zag Polishing Trajectory
  MoveL(Waypoint_2)
  MoveL(Waypoint_3)
  ... ▷ Intermediate Waypoints
  MoveL(Waypoint_14)
End Force Mode
StopL(Deceleration=5.0)

Task parameters

The following parameters were defined and tuned to ensure the successful execution of the polishing task (Application B) using the UR3e robot and the Robotiq E-Pick tool.

Tool Configuration (TCP & Payload)

Based on the Robotiq E-Pick specifications:

- **TCP Position** (X, Y, Z)**:** $(0, 0, 145)$ mm. (Defined at the tip of the suction cup).
- **TCP Orientation** (R, P, Y)**:** $(0, 0, 0)$ rad.
- **Payload Mass:** 0.95 kg.
- **Center of Gravity (CoG):** $(6, -1, 45)$ mm.

Force Control Parameters

Configured within the `force_mode()` script command:

- **Feature Frame:** Tool Frame (Force applied relative to tool orientation).
- **Compliant Axis:** Z-axis only ($[0, 0, 1, 0, 0, 0]$).
- **Target Force:** 10.0 N (Applied in Z+ to press against surface).
- **Damping:** 0.1 (Default damping for stability).

Speeds and Accelerations

- **Global Acceleration:** 1.2 rad/s^2.
- **Fast Approach Speed (MoveL):** 250 mm/s.
- **Surface Search Speed:** 20 mm/s. Critical to detect contact without collision.
- **Polishing Speed:** < 100 mm/s inside Force node to ensure safety.

Safety Limits & I/O

- **Safety Planes:**
  - *Table Surface:* Mode "Normal" (Restricted workspace).
  - *Back Wall:* Mode "Both" (Reduced Speed + Stop).
- **Tool Orientation Limit:** Max deviation of 20° (0.35 rad) from vertical normal.
- **Digital I/O:** DI0 (Start), DO0 (Running Light), DO3 (Alarm Light).

Safety Considerations

Since the UR3e is a collaborative robot sharing the workspace with the operator, strict safety measures were implemented.

1. Workspace Restriction (Safety Planes)

To prevent the robot from colliding with the environment, three safety planes were configured in the PolyScope *Installation* > *Safety* tab. The **Table Surface Plane** was set to Normal Mode to protect mounting equipment, while the **Back Wall Plane** was set to Both Mode (Reduced Speed + Stop) to prevent impacts with the lab structure.

2. Application-Specific Safety (Logic)

Because "Force Mode" changes the robot's behavior, specific logical checks were programmed:

- **Orientation Monitoring:** A parallel thread continuously monitors the tool's inclination using `get_actual_tcp_pose()`. If the tool tilts more than 20°, the program triggers a halt and activates the Alarm Light (DO3).
- **Force Mode Speed Limit:** While in Force Mode, speed is strictly limited to 100 mm/s to prevent dangerous acceleration if resistance is lost.
- **Slow Contact Approach:** The "Surface Detection" phase uses a very low speed (20 mm/s) to prevent high-impact collisions.

Execution Results

The execution of the laboratory session was successful, confirming that the UR3e is capable of performing force-sensitive tasks on irregular surfaces.

- **Sequence Completion:** The program logic functioned correctly. The robot detected the surface height, applied the target force of 10N, and followed the waypoints without errors.
- **Force Compliance:** We visually verified that the robot's Z-axis moved dynamically while traversing the trajectory, confirming the Force Mode was compensating for the curvature.
- **Cycle Time:** The complete cleaning cycle took approximately **45 seconds**.
- **Safety Verification:** The orientation monitor was tested by manually tilting the tool; the robot successfully halted and triggered the alarm.

Improvements

To optimize the task for a real industrial application (cleaning a car door), we propose the following improvements:

1. **Alternative Cleaning Trajectory:** The current "Zig-Zag" pattern is efficient for coverage but can leave linear streaks. We would improve the cleaning method by implementing a **circular or spiral trajectory**. This method mimics human hand polishing, ensuring a better surface finish and preventing directional scratch marks.
2. **Force-Velocity Coupling:** Optimization logic could be added to decrease force slightly if the robot slows down at corners, preventing over-polishing in specific spots.
3. **Sensor-Based Path Planning:** Instead of pre-taught waypoints, the Force/Torque sensor could be used to detect the edges of the door automatically, allowing the robot to clean doors of varying sizes without reprogramming.